Delivery Performance, Delay Risk, and Logistics Efficiency Analysis in Global Supply Chain Operations

Research Paper

Project: Analysis in Global Supply Chain Operations
Eligible shipment records analyzed: 172,765
Generated: 09 November 2026

Abstract

This research examines delivery performance, delay exposure and logistics efficiency using the project's processed supply-chain shipment dataset. The analysis uses 172,765 eligible shipment records after excluding cancelled shipments and records with invalid or missing actual or scheduled shipping durations. Delivery gaps are calculated as actual shipping duration minus scheduled shipping duration. The observed results show that 57.29% of eligible shipments were delayed, 18.64% were exactly on-time, and 24.07% were early. Under the project's SLA-based KPI definition, the On-Time Delivery Rate is 42.71%. The average delay among delayed shipments is 1.62 days. The study compares shipping modes, regions, countries, markets and customer segments, and applies five project-defined logistics KPIs. The findings are descriptive and reflect the fields available in the dataset. The absence of a verified operational date field limits chronological and seasonality analysis. The paper concludes with evidence-based operational recommendations grounded in the observed shipment records.

Introduction

Global supply chains depend on the coordination of transportation, shipment scheduling, regional distribution and customer fulfilment. Delivery performance is an important operational dimension because departures from scheduled shipping durations can affect service reliability, logistics planning and the allocation of operational resources. A structured analysis of actual and scheduled shipping durations can identify the extent of delivery delays and reveal differences across shipping modes, markets, regions and customer segments.

This project examines shipment-level delivery performance using the supplied APL Logistics dataset. It calculates delivery gaps, classifies shipments as delayed, exactly on-time or early, and develops five required logistics KPIs. The analysis is descriptive: it identifies patterns in the available records without claiming that observed associations establish causal relationships.

Background and Context

Logistics performance measurement commonly distinguishes the scheduled duration of a shipment from the duration actually recorded. The difference provides a direct measure of whether a shipment took longer than planned, matched the schedule or arrived earlier than the scheduled duration. These classifications support comparisons between shipping modes and geographic groupings.

In this project, SLA compliance is defined as a delivery gap less than or equal to zero. This includes both exactly on-time and early shipments. Exact on-time delivery is separately defined as a gap equal to zero. This distinction is necessary because an SLA-compliant percentage should not be interpreted as the percentage of shipments with an exact zero-day gap.

Problem Statement

The project addresses the need for a consistent, evidence-based assessment of delivery performance across a large set of logistics records. Without a common delivery-gap calculation and comparable performance indicators, it is difficult to distinguish delayed shipments from early and exactly on-time shipments or compare operational performance across shipping modes, regions, markets and customer segments.

The analysis also needs to account for cancelled shipments and invalid duration records, preserve valid zero-duration Same Day shipments, and avoid unsupported chronological or geographic interpretations.

Project Objectives

- Assess the quality and structure of the processed logistics dataset.
- Calculate delivery gaps using actual and scheduled shipping durations.

- Classify eligible shipments as delayed, exactly on-time or early.
- Measure overall delivery performance and the existing late delivery risk label.
- Compare delivery performance and efficiency across shipping modes.
- Evaluate regional, country and market-level delivery differences.
- Assess delivery performance across Consumer, Corporate and Home Office segments.
- Calculate the five required project logistics KPIs.
- Identify observed operational bottlenecks and formulate evidence-based recommendations.

Dataset Description

The cleaned dataset contains 180,519 records and 32 columns. The delivery-gap analysis dataset contains 180,519 records and 41 columns. The cleaned dataset is stored at `data/processed/cleaned_logistics_data.csv`, while the delivery-gap analysis dataset is stored at `data/processed/delivery_gap_analysis.csv`.

The analytical fields used include actual shipping duration, scheduled shipping duration, delivery status, late delivery risk, shipping mode, order region, order country, market and customer segment. The primary unit of analysis is the available shipment record. The processed dataset covers 23 regions, 5 markets, 164 countries, 4 shipping modes and 3 customer segments.

Dataset Summary

Dataset Characteristic	Value
Cleaned dataset records	180,519
Cleaned dataset columns	32
Delivery gap dataset records	180,519
Eligible shipments	172,765
Regions	23
Markets	5
Countries	164
Shipping modes	4
Customer segments	3
Verified operational date column	Not available

Dataset Limitations

- The processed dataset does not contain a verified operational date column; chronological trend and seasonality analysis cannot be established.
- No synthetic dates have been created.
- The dataset does not provide a dedicated Order ID for order-level deduplication.
- Customer coordinates are not treated as confirmed shipment destination coordinates.
- Late_delivery_risk is an existing binary dataset label, not a newly trained probability model.
- Customer segments are interpreted as Consumer, Corporate and Home Office; Corporate is not assumed to mean premium or enterprise.
- The analysis is based on shipment records and available fields; causal explanations cannot be established from these descriptive associations alone.
- The Shipping Mode Efficiency Index and Regional Delay Index are project-defined analytical measures, not externally standardized industry benchmarks.

Data Cleaning Methodology

The research uses the project's processed logistics datasets. The cleaned dataset contains 180,519 records. For delivery performance calculations, actual and scheduled durations are converted to numeric values. Records are considered eligible when both duration values are present, non-negative and the delivery gap is valid. Cancelled shipments are excluded from the eligible performance population.

Valid zero scheduled-duration records are retained. A zero scheduled duration is not automatically treated as an error because the dataset contains Same Day shipments. Missing or invalid durations are not converted to zero. Personal customer identifiers were excluded from the cleaned analytical dataset. The analysis does not deduplicate customer IDs as if they were order IDs.

Delivery Gap Calculation Methodology

The delivery gap is calculated using the following formula:

Delivery Gap = Actual Shipping Duration - Scheduled Shipping Duration

A positive value indicates that actual shipping took longer than scheduled and the shipment is classified as delayed. A value of zero indicates an exactly on-time shipment. A negative value indicates an early shipment.

For the project's On-Time Delivery Rate KPI, SLA compliance is calculated as the number of eligible shipments with a delivery gap less than or equal to zero divided by all eligible shipments, multiplied by 100. This project-defined KPI includes early shipments. Exact on-time performance is separately calculated using a delivery gap equal to zero.

Delivery Performance Analysis

The eligible population contains 172,765 shipments. Among these records, 98,977 shipments were delayed, 32,196 were exactly on-time and 41,592 were early. These categories reconcile to the eligible shipment population. The SLA-compliant population contains 73,788 shipments because both exactly on-time and early shipments satisfy the project's gap less than or equal to zero definition.

The delayed-shipment percentage is 57.29%. The exact on-time percentage is 18.64%, while the early shipment percentage is 24.07%. The project's SLA-based On-Time Delivery Rate KPI is 42.71%. The mean positive delivery gap among delayed shipments is 1.62 days and the median positive delay is 1.00 days.

Delivery Classification Results

Delivery Classification	Shipment Count	Percentage
Delayed	98,977	57.29
Exactly On-time	32,196	18.64
Early	41,592	24.07
SLA-compliant	73,788	42.71

Shipping Mode Efficiency Analysis

Shipping-mode analysis compares shipment volume, delayed shipment percentage, exact on-time percentage, SLA compliance, average positive delay and the project-defined Shipping Mode Efficiency Index. The highest-ranked mode by the calculated efficiency index is Standard Class. The largest shipment volume is associated with Standard Class. The highest average positive delay is observed for Second Class.

The efficiency index combines three scaled components: 50% SLA compliance, 30% inverse late delivery risk and 20% relative delay-gap score. The index is intended for comparison within this project and should not be interpreted as an externally standardized industry benchmark.

Shipping Mode Performance

Efficiency Rank	Shipping Mode	Shipment Count	Delayed Shipments	Exactly On-time Shipments	Early Shipments	SLA-compliant Shipments	Delayed Shipment Percentage	Exact On-time SLA Compliance	Average Recorded Delay Days	Risk Percentage	Shipping M
1	Standard Class	103,153	41,023	20,538	41,592	62,130	39.77	19.91	60.23	1.51	39.77	56.11
2	Same Day	9,293	4,454	4,839	0	4,839	47.93	52.07	52.07	1.00	47.93	53.65
3	Second Class	33,806	26,987	6,819	0	6,819	79.83	20.17	20.17	2.50	79.83	16.14
4	First Class	26,513	26,513	0	0	0	100.00	0.00	0.00	1.00	100.00	11.99

Regional and Market Analysis

Regional analysis compares shipment volume, delayed-shipment percentage, SLA compliance and average positive delivery delay. Central America has the largest shipment volume in the regional comparison. Central Asia has the highest average positive delay. The Regional Delay Index scales each region's average positive delay against the maximum regional average positive delay, with the maximum set to 100.

Market-level analysis identifies LATAM as the largest market by shipment volume and Europe as the market with the highest delayed-shipment percentage. Country-level comparisons are descriptive and depend on the number of records available for each country. These comparisons do not establish why a particular geography experiences a higher delay percentage.

Regional Performance

Order Region	Shipment Count	Delayed Shipments	Exactly On-time Shipments	Early Shipments	SLA-compliant Shipments	Delayed Percentage	SLA Compliance Percentage	Average Delay Days	Regional Delay
Central Africa	1,616	972	290	354	644	60.15	39.85	1.61	95.73
Western Europe	25,867	15,140	4,589	6,138	10,727	58.53	41.47	1.63	97.08
South of USA	3,855	2,256	672	927	1,599	58.52	41.48	1.65	98.38
South Asia	7,455	4,350	1,340	1,765	3,105	58.35	41.65	1.63	97.14
East Africa	1,781	1,036	346	399	745	58.17	41.83	1.60	95.05
East of USA	6,617	3,849	1,208	1,560	2,768	58.17	41.83	1.59	94.95
Southeast Asia	9,136	5,297	1,631	2,208	3,839	57.98	42.02	1.60	95.22
West Asia	5,746	3,322	1,114	1,310	2,424	57.81	42.19	1.60	95.27
Eastern Europe	3,785	2,182	724	879	1,603	57.65	42.35	1.63	96.91
US Center	5,653	3,252	1,174	1,227	2,401	57.53	42.47	1.59	94.81
South America	14,184	8,111	2,597	3,476	6,073	57.18	42.82	1.62	96.45
Central America	27,174	15,518	5,090	6,566	11,656	57.11	42.89	1.61	96.09
North Africa	3,086	1,762	564	760	1,324	57.10	42.90	1.60	95.42
Southern Europe	9,030	5,129	1,655	2,246	3,901	56.80	43.20	1.56	92.92
West of USA	7,595	4,313	1,447	1,835	3,282	56.79	43.21	1.64	97.86
Eastern Asia	6,973	3,955	1,264	1,754	3,018	56.72	43.28	1.66	98.55
Central Asia	542	306	124	112	236	56.46	43.54	1.68	100.00
Oceania	9,733	5,482	1,927	2,324	4,251	56.32	43.68	1.62	96.47
Northern Europe	9,408	5,292	1,775	2,341	4,116	56.25	43.75	1.63	97.25
Southern Africa	1,100	617	200	283	483	56.09	43.91	1.54	91.47
Caribbean	7,951	4,415	1,539	1,997	3,536	55.53	44.47	1.63	96.87
West Africa	3,571	1,953	769	849	1,618	54.69	45.31	1.63	97.06
Canada	907	468	157	282	439	51.60	48.40	1.63	96.80

Market Performance

Market	Shipment Count	Delayed Shipments	Exactly On-time Shipments	Early Shipments	SLA-compliant Shipments	Delayed Percentage	SLA Compliance Percentage	Average Delay Da
Europe	48,090	27,743	8,743	11,604	20,347	57.69	42.31	1.62
USCA	24,627	14,138	4,658	5,831	10,489	57.41	42.59	1.62
Pacific Asia	39,585	22,712	7,400	9,473	16,873	57.38	42.62	1.62
LATAM	49,309	28,044	9,226	12,039	21,265	56.87	43.13	1.62
Africa	11,154	6,340	2,169	2,645	4,814	56.84	43.16	1.60

Country Performance - Highest Delay Percentages

Order Country	Shipment Count	Delayed Shipments	Delayed Percentage	Average Delay Days
Eritrea	2	2	100.00	2.00
Bután	5	5	100.00	2.00
Laos	6	6	100.00	1.83
Guinea Ecuatorial	2	2	100.00	1.50
República de Gambia	5	5	100.00	2.00
Suazilandia	5	5	100.00	2.00
Sudán del Sur	5	5	100.00	2.20
Sáhara Occidental	2	2	100.00	2.00
Luxemburgo	10	10	100.00	2.70
Armenia	5	5	100.00	1.00
Estonia	29	28	96.55	1.61
República del Congo	13	12	92.31	1.25
Eslovenia	6	5	83.33	2.20
Guyana	19	15	78.95	1.60
Gabón	45	35	77.78	1.54

Customer Segment Impact Analysis

The customer-segment analysis compares Consumer, Corporate and Home Office records using shipment volume, delayed-shipment percentage, exact on-time percentage, SLA compliance, average positive delay and the available late delivery risk label.

Consumer has the largest shipment volume, while Home Office has the highest delayed-shipment percentage in the segment comparison. Segment labels are interpreted exactly as provided in the dataset. Corporate is not treated as a premium or enterprise classification.

Customer Segment Performance

Customer Segment	Shipment Count	Delayed Shipments	Exactly On-time Shipments	Early Shipments	SLA-compliant Shipments	Delayed Percentage	Exact On-time Percentage	SLA Compliance Percentage	Average Delay Days	Risk Perc
Home Office	30,817	17,747	5,722	7,348	13,070	57.59	18.57	42.41	1.63	57.59
Consumer	89,420	51,248	16,784	21,388	38,172	57.31	18.77	42.69	1.61	57.31
Corporate	52,528	29,982	9,690	12,856	22,546	57.08	18.45	42.92	1.62	57.08

KPI Methodology

The five required KPIs are calculated as follows:

On-Time Delivery Rate (%) = SLA-compliant shipments / eligible shipments x 100. SLA-compliant shipments have a delivery gap less than or equal to zero.

Average Delivery Delay (Days) = Mean of positive delivery gaps among delayed eligible shipments only.

Late Delivery Risk Ratio (%) = Mean of valid binary Late_delivery_risk labels multiplied by 100.

Shipping Mode Efficiency Index = 50% SLA compliance score + 30% inverse late-risk score + 20% relative delay-gap score. The three components are expressed on a 0-100 scale.

Regional Delay Index = Regional average positive delay / Maximum regional average positive delay x 100. The region with the maximum average positive delay receives an index of 100.

These definitions are the project's specified KPI formulas. The Shipping Mode Efficiency Index and Regional Delay Index are project-defined comparative indicators.

Five Required Logistics KPIs

KPI	Value	Calculation
On-Time Delivery Rate (%)	42.71	SLA-compliant shipments / eligible shipments x 100
Average Delivery Delay (Days)	1.62	Mean positive delay gap among delayed shipments
Late Delivery Risk Ratio (%)	57.29	Mean valid binary Late_delivery_risk label x 100
Shipping Mode Efficiency Index	56.11	50% SLA compliance + 30% inverse risk + 20% delay score
Regional Delay Index	100.00	Regional average positive delay / maximum regional average positive de

Results and Findings

- The analysis contains 172,765 eligible shipment records after excluding cancelled shipments and records with invalid or missing duration values.
- 98,977 eligible shipments (57.29%) have a positive delivery gap and are classified as delayed.
- 32,196 shipments (18.64%) have a zero delivery gap and are exactly on-time.
- 41,592 shipments (24.07%) have a negative delivery gap and are classified as early.
- 73,788 shipments (42.71%) are SLA-compliant under the project's definition of a gap less than or equal to zero.
- The average delay among delayed shipments is 1.62 days, while the median positive delay is 1.00 days.
- The existing Late_delivery_risk label has a ratio of 57.29% among valid binary risk records.
- Standard Class has the largest shipment volume among shipping modes.
- Standard Class has the highest project-defined Shipping Mode Efficiency Index.
- Second Class has the highest average positive delay among shipping modes.
- Central America has the largest shipment volume among regions.
- Central Asia has the highest regional average positive delay.
- LATAM has the largest shipment volume among markets.
- Europe has the highest delayed-shipment percentage among markets.
- Consumer has the largest shipment volume among customer segments.
- Home Office has the highest delayed-shipment percentage among customer segments.
- Eritrea has the highest delayed-shipment percentage in the country-level comparison.

Operational Bottlenecks

- Overall delayed-shipment exposure is 57.29% of eligible shipments.
- The shipping mode with the highest average positive delay is Second Class.
- The region with the highest average positive delay is Central Asia.
- The market with the highest delayed-shipment percentage is Europe.
- The customer segment with the highest delayed-shipment percentage is Home Office.
- The absence of a genuine operational date column restricts chronological monitoring, trend analysis and seasonality assessment.

Business Recommendations

1. Prioritize operational review of the shipping mode with the highest average positive delay and compare its shipment volume, SLA compliance and existing risk-label distribution.
2. Investigate the regions and markets with the highest observed delay measures to identify operational causes using additional verified logistics records.
3. Use the exact on-time, early and delayed classifications separately in performance reporting so that early deliveries are not confused with exactly on-time deliveries.
4. Monitor SLA compliance and average delay together. A single percentage does not describe the magnitude of delays among shipments that miss their scheduled duration.
5. Review the existing Late_delivery_risk label against observed delivery outcomes before using it as an operational decision signal.
6. Assess customer segments using their observed shipment counts, delay percentages and SLA compliance without assigning unsupported premium or enterprise categories.
7. Collect and validate genuine order, dispatch and delivery timestamps in future data releases to enable chronological forecasting, trend analysis and date-based operational monitoring.
8. Maintain consistent shipment-level data quality checks, including validation of actual and scheduled durations and clear treatment of cancelled shipments.

Conclusion

The project provides a structured descriptive assessment of delivery performance using 172,765 eligible shipment records. The delivery-gap method distinguishes delayed, exactly on-time and early shipments, while the project-defined SLA compliance KPI reports a value of 42.71%. The average positive delay among delayed shipments is 1.62 days.

Shipping-mode, regional, market and customer-segment comparisons identify differences in shipment volume and observed delivery performance. The five required KPIs provide a consistent framework for comparing operational outcomes. The findings support targeted review of the modes and groups with higher observed delay measures.

The results must be interpreted within the limitations of the available data. In particular, the absence of a genuine operational date field prevents chronological trend and seasonality analysis. Further work would benefit from validated order and delivery timestamps and additional operational variables that can help investigate the causes of delay.

References

1. APL Logistics dataset supplied for the project, original workbook: data/raw/APL_Logistics.xlsx.
2. Project processed dataset: data/processed/cleaned_logistics_data.csv.
3. Project delivery-gap analysis dataset: data/processed/delivery_gap_analysis.csv.
4. Project analysis outputs: shipping mode, regional, market, customer-segment and KPI analysis tables generated from the supplied dataset.

5. Pandas documentation. https://pandas.pydata.org/docs/.

6. NumPy documentation. https://numpy.org/doc/.

7. Plotly Python documentation. https://plotly.com/python/.

8. Streamlit documentation. https://docs.streamlit.io/.